

Jump-Diffusion Models for Financial Markets

Alex Faassen

December 16, 2024

Abstract

This report investigates the foundation, methodology, and applications of Jump-Diffusion models. While the Black-Scholes model is widely valued for its simplicity and versatility, it falls short in capturing the heavier tails and higher peaks observed in most financial asset returns. To address these limitations, among others, Robert C. Merton extended the Black-Scholes model in 1976 by incorporating a Compound Poisson Process to account for random jumps in asset prices. By simulating asset prices and analyzing option pricing under the Merton and Kou Jump-Diffusion models alongside the Black-Scholes model, this report offers a practical introduction to their mechanics, interpretations, and comparative advantages.

1 Introduction

In class, the study of the necessary tools provided a deeper understanding of the derivation of the Black-Scholes model and its European option pricing formulas. Celebrated for its versatility and simplicity, the Black-Scholes model marks a historic milestone in financial mathematics. However, it leaves many features of financial asset prices unaddressed. This report examines Jump-Diffusion models, which extend the Black-Scholes framework to account for some of these shortcomings.

What are Jump-Diffusion models, and how do they work? How do they compare to the Black-Scholes model? To answer these questions, this report begins by exploring the historical context behind Jump-Diffusion models to understand their purpose and innovation. Familiar tools are extended, introducing Compound Poisson Processes, to build a foundation for understanding modifications to the Black-Scholes model.

A general definition of Jump-Diffusion models enables comparisons between two prominent variations: the Merton and Kou models. This report examines their interpretations and applications. To solidify this understanding, simulated asset price paths are analyzed under different parameters across the Black-Scholes, Merton, and Kou models. Additionally, a brief discussion of their European Option Pricing formulas provides deeper insight into their deviations. Finally, with a renewed understanding of Jump-Diffusion models, the findings are summarized to highlight the comparative strengths of the Black-Scholes and Jump-Diffusion models.

2 History

In 1976, Robert C. Merton extended the Black-Scholes model by introducing a jump term. Before this development, Dr. Merton had already made significant contributions to the foundational theory behind the Black-Scholes model, often referred to as the Black-Scholes-Merton model. This pioneering work on Jump-Diffusion models addressed the empirically observed heavier tails and higher peaks of financial asset returns, which the standard Normal distribution fails to capture. Moreover, it was later recognized that incorporating jump terms also helped explain phenomena such as the Volatility Smile.

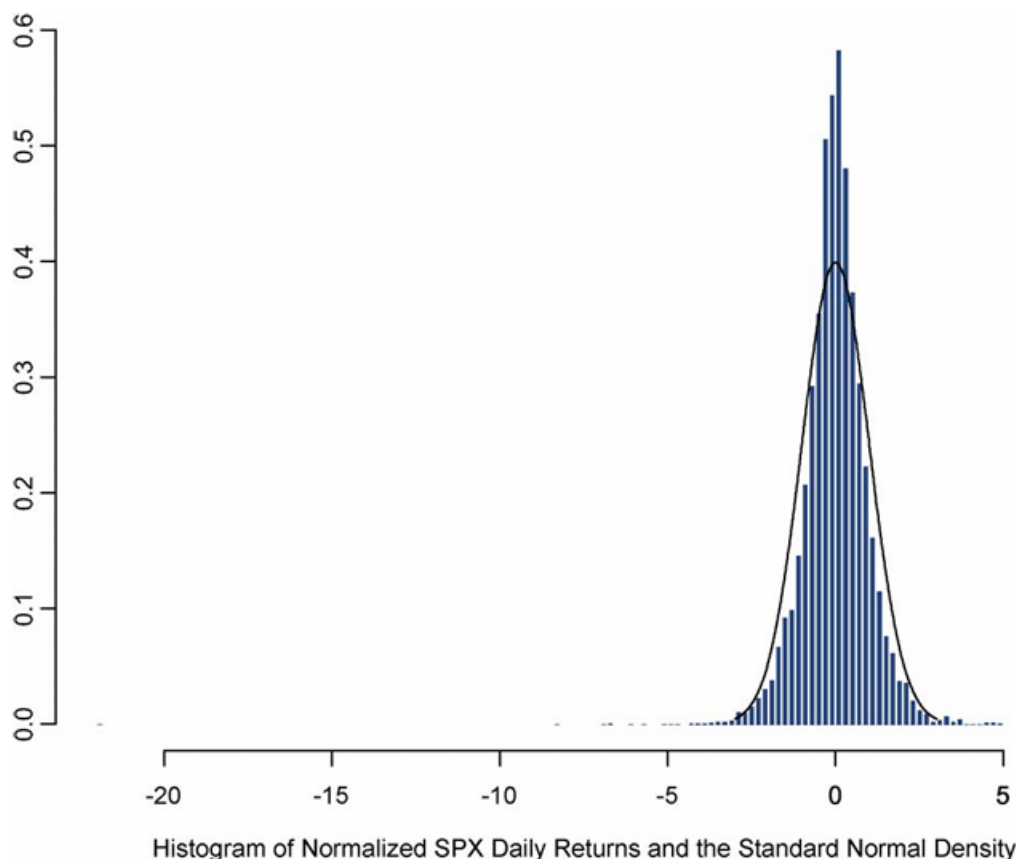


Figure 1: Comparison of the histogram of the normalized daily returns of S&P 500 index (from Jan 2, 1980 to Dec 31, 2005) and the density of $N(0, 1)$. The feature of a high peak and two heavy tails (i.e. the leptokurtic feature) is quite evident. Reprinted from [Kou07].

2.1 Leptokurtic Distribution

Figure 1 illustrates the normalized returns of the S&P 500, which exhibit heavier tails and a higher peak than the overlaid standard Normal distribution, $N(0, 1)$. These features have been empirically observed for most financial assets [Kou07], characterizing a larger kurtosis than the standard Normal distribution.

Kurtosis measures the shape of a distribution, particularly the weight of its tails. Distributions with higher kurtosis than the standard Normal distribution, which has a kurtosis of 3, are described as **leptokurtic distributions**, meaning they exhibit "large kurtosis." However, the Black-Scholes model assumes Normally distributed increments translating to returns which therefore fail to reflect the characteristics of leptokurtic distributions.

2.2 Volatility Smile

Assuming we know the parameters in a Black-Scholes European Option pricing formula, we can define an inverse function mapping from a given option price to the volatility parameter since the Black-Scholes formulas are monotonically increasing with respect to diffusion. The representation of this mapping as a function of time to maturity and strike price is referred to as **implied volatility**, $\sigma(T, K)$.

Implied volatilities can be used to price options with different strike prices and maturity times, removing the need for historical data. Accordingly, this can be useful when historical volatilities do not reflect current market conditions well. In practice, plotting implied volatilities against strike prices produces a convex curve

resembling a "volatility smile". However, the Black-Scholes model utilizes Geometric Brownian Motion which assumes implied volatilities are the same for all options with the same underlying asset, therefore failing to reflect the volatility smile.

3 Poisson Processes

Jump-Diffusion models address these features with the addition of a jump term to the Black-Scholes model. "Jumps" refer to random, abrupt changes in the underlying asset price. To model jumps, several dimensions must be considered, including the number of jumps, the timing of these jumps, and their magnitudes. The concept of Compound Poisson Processes is explored to represent these quantities mathematically.

Poisson Process: a stochastic process that counts the number of random events occurring within a specified period. These events are modelled by exponential random variables, with the process defined as:

$$N_t = \sum_{n \geq 1} 1_{t \geq T_n},$$

where $T_n = \sum_{i=1}^n \tau_i$ and $(\tau_i)_{i \geq 1}$ are independent and identically distributed (i.i.d.) exponential random variables with parameter λ , $(\tau_i)_{i \geq 1}$ i.i.d. *Exponential*(λ). λ is called the **intensity** of N_t , which determines the average frequency of jumps over time.

In essence, Poisson processes provide a clear representation of the relationship between Poisson and Exponential random variables. A Poisson Process of intensity λ can be used to model the number of jumps and their timing up to the maturity time of an option.

Compound Poisson Process: using N_t as the number of jumps and another independent sequence of i.i.d. RVs $(Y_i)_{i \geq 1}$ to model jump sizes, the process is defined as:

$$X_t = \sum_{i=1}^{N_t} Y_i$$

By introducing a third set of random variables, the Compound Poisson Process can model not only the number and timing of jumps (governed by the Poisson Process) but also their magnitudes (modelled by Y_i). This approach allows for flexibility in adjusting the jump size distribution while maintaining the desired intensity, λ , of the Poisson Process.

4 Jump-Diffusion Models

The Black-Scholes model describes the dynamics of two assets using the following stochastic differential equations (SDEs):

$$\begin{aligned} dB_t &= rB_t dt \\ dS_t &= \mu S_t dt + \sigma S_t dW_t \end{aligned}$$

where r , μ , and σ are constants and W_t represents standard Brownian Motion.

To incorporate sudden, random changes (jumps) in the asset price, a Compound Poisson Process term is added to the dynamics of the stock price S_t . This adjustment yields the general form of Jump-Diffusion models.

Jump-Diffusion Model:

$$dB_t = rB_t dt$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t \sum_{i=1}^{N_t} (Y_i - 1)$$

where N_t is a Poisson process with intensity λ and $(Y_i)_{i \geq 1}$ is a sequence of i.i.d. nonnegative random variables representing jump sizes.

The term $Y_i - 1$ reflects the assumption that after a jump, the stock price transitions as $S_t \rightarrow S_t * Y_t$. In this case, the size of the jump can be determined by $S_t Y_t - S_t = S_t(Y_t - 1)$.

To derive the dynamics of the asset price, Proposition 8.14 of Cont and Tankov (2004) [CT04] gives the Itô formula for jump-diffusion processes. In a more familiar format, this is:

Let the jump-diffusion process X have SDE $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t + \sum_{i=1}^{dN_t} (Y_i - 1)$, where μ, σ are adapted, integrable processes. Let f be a $C^{1,2}$ function. Then:

$$d(f(t, X_t)) = \left[\frac{\delta f}{\delta t}(t, X_t) + \mu(t, X_t) \frac{\delta f}{\delta X}(t, X_t) + \frac{1}{2} \sigma^2(t, X_t) \frac{\delta^2 f}{\delta x^2}(t, X_t) \right] dt + \sigma(t, X_t) \frac{\delta f}{\delta X}(t, X_t) dW_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})]$$

where X_{t-} refers to the value of X_t when approaching t from the left, in reference to discontinuities created by jumps.

Applying this version of Itô's Lemma to solve the above SDE provides a formula for the asset price under Jump-Diffusion models.

$$\begin{aligned} d\ln(S_t) &= \left[0 + \mu S_t \frac{1}{S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{-1}{S_t^2} \right] dt + \sigma S_t \frac{1}{S_t} dW_t + [\ln(S_t + S_t(Y_t - 1)) - \ln(S_t)] \\ \rightarrow d\ln(S_t) &= \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t + [\ln(S_t Y_t) - \ln(S_t)] \\ \rightarrow d\ln(S_t) &= \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t + \ln(Y_t) \\ \rightarrow \ln(S_t) - \ln(S_0) &= \int_0^t \left(\mu - \frac{\sigma^2}{2} \right) ds + \int_0^t \sigma dW_u + \sum_{i=1}^{N_t} \ln(Y_i) \\ \rightarrow \ln(S_t) &= \ln(S_0) + \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t + \sum_{i=1}^{N_t} \ln(Y_i) \end{aligned}$$

Thus, we have the following, similar to the asset price dynamics under Geometric Brownian Motion.

$$S_t = S_0 \exp\left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right\} \prod_{i=1}^{N_t} Y_i$$

4.1 Merton vs. Kou Models

Two notable Jump-Diffusion models are the Merton model and the Kou model. The Merton model being the first Jump-Diffusion model that Dr. Merton described in his 1976 paper. The Kou model was later introduced in a 2002 paper authored by Steven G. Kou. This model attempted to build upon the Merton model, providing solutions for a broader range of financial instruments, including path-dependent options like barrier and lookback options.

The primary distinction between these two models lies in the distribution of jump sizes, Y_i [Kou07]:

- In the **Merton model**, $\log(Y_i)$ is normally distributed, $N(m, \sigma^2)$, making Y_i lognormal, a more natural extension of the Black-Scholes model.

- In the **Kou model**, $\log(Y_i)$ follows an asymmetric double exponential distribution, allowing for distinct modelling of positive and negative jumps.

The probability density function (pdf) for jump sizes in the Kou model is given by:

$$f(x) = p\eta_1 e^{-\eta_1 x} 1_{x \geq 0} + (1-p)\eta_2 e^{-\eta_2 x} 1_{x < 0},$$

where p is the probability of a positive jump and η_1, η_2 are rate parameters for positive and negative jumps respectively.

These differences lead to simpler derivations for the Merton model and greater versatility with the Kou model. Table 1 summarizes the key parameters of the two models [CT04].

Merton (4 parameters)	Kou (5 parameters)
σ - diffusion (volatility), λ - jump intensity (frequency), m - mean jump size, and δ - jump size standard deviation.	σ - diffusion (volatility), λ - jump intensity (frequency), p - probability of positive (vs. negative) jump, and $\frac{1}{\eta_1}, \frac{1}{\eta_2}$ - mean positive/negative jump sizes.

Table 1: Parameters for the Merton and Kou Jump-Diffusion models.

5 Simulation

With the definitions of the Merton and Kou models established, simulations can be conducted by combining the standard Brownian motion and Compound Poisson processes. Since these components are independent, a discretized trajectory can be constructed by simulating each process separately and combining them, as outlined in Algorithm 6.3 of Cont and Tankov (2004) [CT04].

$$S_{t_i} = \mu t_i + \sum_{k=1}^i G_k + \sum_{j=1}^N 1_{U_j < t_i} Y_j$$

for $t_1, \dots, t_n$, where μ, σ are constants, (G_i) are i.i.d. Normal with $\text{Var}(G_i) = (t_i - t_{i-1})\sigma^2$, $N \sim \text{Poisson}(\lambda)$, (U_i) are i.i.d. $\text{Uniform}[0, T]$, and (Y_i) are i.i.d. according to the model used.

5.1 Sample Paths

Simulated asset price trajectories of the Black-Scholes, Merton, and Kou models using the above algorithm with constant drift (μ) and diffusion (σ) parameters illustrate their key differences. Starting with modest and then moving to larger jump parameters highlights the effect of the jump terms compared to the Black-Scholes model.

Figure 2 provides a side-by-side view of summarized sample path simulations for the Black-Scholes, Merton, and Kou models on a fixed y-axis. The black lines reflect the 10th, 50th, and 90th percentiles at each time t , while the coloured lines are 10 randomly chosen simulations. All 3 models are simulated with constant Brownian motion drift and diffusion parameters, $\mu = 0.05$ and $\sigma = 0.5$. The Merton and Kou model jump terms are kept modest, at $m = 0.01$, $\delta = 0.05$, $p = 0.5$, and $\eta_1 = \eta_2 = 10$ respectively. On the left, the Black-Scholes model shows smooth trajectories driven by Brownian motion. The accompanying Jump-Diffusion models appear fairly similar to Brownian motion. Of course, this is expected given that they are additive models.

Figure 3 utilizes the same format as Figure 2. However, the jump parameters for the Merton and Kou models are significantly larger, at $m = 0.05$, $\delta = 0.25$, $p = 0.7$, and $\eta_1 = \eta_2 = 5$ respectively. Immediately, greater jump parameters seem to produce significantly different plots from that of the Black-Scholes model with some of the randomly chosen sample paths reflecting some rather large jumps. Both of the Merton and

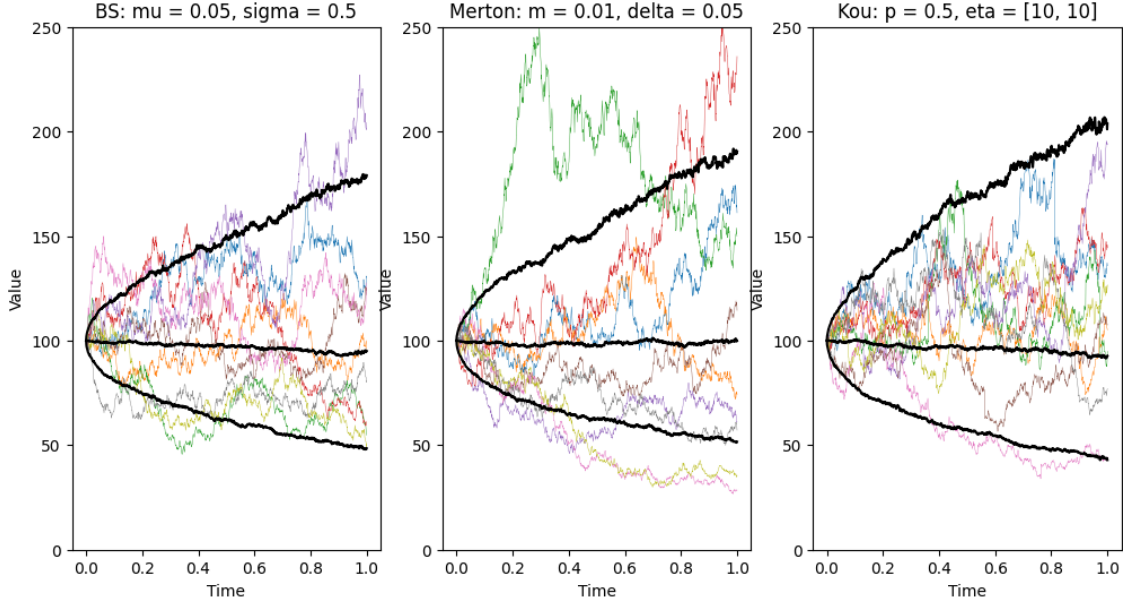


Figure 2: Simulated sample paths for the Black-Scholes (standard Brownian motion), Merton (Jump-Diffusion), and Kou models. Constant $\mu = 0.05$ and $\sigma = 0.5$, but differing jump parameters. Merton: $m = 0.01$, $\delta = 0.05$. Kou: $p = 0.5$, $\eta_1 = \eta_2 = 10$. On 1000 simulated sample paths, the black lines represent the 10th, 50th, and 90th percentiles at each time, and the coloured lines are 10 randomly chosen sample paths.

Kou model plots display significantly greater volatility than that of the Black-Scholes model. Once again, this is expected from an additive model with constant Brownian motion parameters.

5.2 Intuition

Understanding the intuition behind these models can guide their application to real-world financial problems. Dr. Kou emphasized in his paper *Chapter 2: Jump-Diffusion Models for Asset Pricing in Financial Engineering* [Kou07] that jumps could be interpreted as reflecting the reality of sudden market movements, such as news events or economic shocks. In the context of a Jump-Diffusion model, this means that without any external information, the asset price follows a Geometric Brownian Motion. However, external market information arrives according to a Poisson process. Dr. Kou suggests that "the double-exponential jump-diffusion model can be interpreted as an attempt to build a simple model, within the traditional random walk and efficient market framework, to incorporate investors' sentiment." [Kou07] Since the Merton model shares a near identical framework with the Kou model, it follows that the same intuition can likely be applied to the Merton model and Jump-Diffusion models in general. This contrasts with the smooth evolution assumed by Brownian motion in the Black-Scholes model.

6 Option Pricing

European put/call option pricing offers valuable insights into the differences between Jump-Diffusion models and the Black-Scholes model. This section focuses on the Merton model option pricing formula as it should suffice for understanding fundamental differences. It should be noted that the Kou model also has a closed-form formula and approximations via Laplace transformations for European put/call options. However, the formulas and their derivations quickly become more complex.

Below, the Jump-Diffusion model is defined like the Black-Scholes model in class.

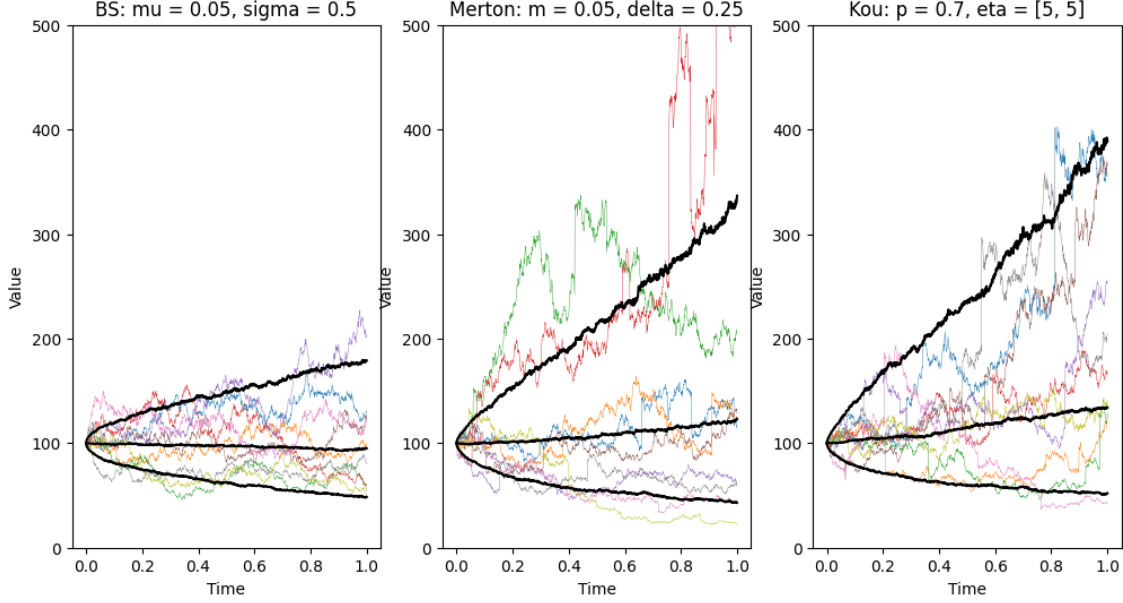


Figure 3: Simulated sample paths for the Black-Scholes (standard Brownian motion), Merton (Jump-Diffusion), and Kou models. Constant $\mu = 0.05$ and $\sigma = 0.5$, but differing jump parameters. Merton: $m = 0.05$, $\delta = 0.25$. Kou: $p = 0.7$, $\eta_1 = \eta_2 = 5$. On 1000 simulated sample paths, the black lines represent the 10th, 50th, and 90th percentiles at each time, and the coloured lines are 10 randomly chosen sample paths.

Jump-Diffusion Model: The Jump-Diffusion model consists of two assets with dynamics:

$$dB_t = rB_t dt, \quad B_0 = 1$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t \sum_{i=1}^{dN_t} (Y_i - 1), \quad S_0 = s_0$$

Since this model is incomplete, there are many possible choices for a risk-neutral measure, Q , a set of dynamics under which the discounted price, $e^{-rt}S_t$, is a martingale [CT04]. For simplicity, Merton chose to emulate the Black-Scholes model, changing the drift of the Brownian motion, μ , leaving everything else unchanged.

It can be shown that with the choice of drift parameter [Kou07] [Mat04], $\mu^Q = r - \frac{\sigma^2}{2} - \lambda(E^Q[e^Y] - 1)$, $S_t e^{-rt}$ is a martingale under Q dynamics:

$$dS_t = \left(r - \frac{\sigma^2}{2} - \lambda(E^Q[e^Y] - 1)\right) S_t dt + \sigma S_t dW_t^Q + S_t \sum_{i=1}^{dN_t} (Y_i - 1)$$

$$S_t = s$$

where W^Q is a standard Brownian motion under Q and $E^Q[e^Y] = e^{m + \frac{\delta^2}{2}}$, since $Y \sim \text{Lognormal}(m, \delta^2)$.

Solving for $f(t, s) = e^{-r(T-t)} E^Q[F(S_T) | F_t]$ [CT04] [Mat04], gives a Merton model formula.

$$\begin{aligned}
f(t, s) &= e^{-r(T-t)} E^Q[F(S_T)|F_t] = e^{-r(T-t)} E^Q[(S_T - K)_+ | S_t = s] \\
&= e^{-r\Delta t} E^Q[F(\exp\{\mu^Q \Delta t + \sigma W_{\Delta t}^Q + \sum_{i=1}^{N_{\Delta t}} \log(Y_i)\})] \\
&= e^{-r\Delta t} \sum_{n \geq 0} Q(N_t = n) E^Q[F(\exp\{(r - \frac{\sigma^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t + \sigma W_{\Delta t}^Q + \sum_{i=1}^n Y_i\})]
\end{aligned}$$

where $(r - \frac{\sigma^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t + \sigma W_{\Delta t}^Q + \sum_{i=1}^n Y_i$ and $(r - \frac{\sigma^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t + nm + \sqrt{\frac{\sigma^2 \Delta t + n\delta^2}{\Delta t}} W_{\Delta t}^Q$ are both $N((r - \frac{\sigma^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t + nm, \sigma^2 \Delta t + n\delta^2)$, which is substituted for simplification.

$$\begin{aligned}
&= e^{-r\Delta t} \sum_{n \geq 0} \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^n}{n!} E^Q[F(\exp\{(r - \frac{\sigma^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t + nm + \sqrt{\frac{\sigma^2 \Delta t + n\delta^2}{\Delta t}} W_{\Delta t}^Q\})] \\
&= \sum_{n \geq 0} \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^n}{n!} e^{-r\Delta t} E^Q[F(\exp\{nm + \frac{n\delta^2}{2} - \lambda(e^{m+\frac{\delta^2}{2}} - 1))\Delta t\} \exp\{(r - \frac{\sigma^2}{2})\Delta t + \sigma_n W_{\Delta t}^Q\})]
\end{aligned}$$

Thus,

$$f_{Merton}(t, S_t) = \sum_{n=0}^{\infty} \frac{e^{-\bar{\lambda} \Delta t} (\bar{\lambda} \Delta t)^n}{n!} f_{BS}(t, S_t, \sigma = \sigma_n, r = r_n),$$

where $\bar{\lambda} = \lambda e^{m+\frac{\delta^2}{2}}$, $\sigma_n = \sqrt{\sigma^2 + \frac{n\delta^2}{\Delta t}}$, and $r_n = r - \lambda(e^{m+\frac{\delta^2}{2}} - 1) + \frac{nm + \frac{n\delta^2}{2}}{\Delta t}$.

Interestingly, this formula can be interpreted as a weighted average of Black-Scholes prices conditioned on the number of jumps, n .

Note: Put-Call Parity is model-agnostic. Hence, European Put Option pricing can still be derived from Call Option pricing.

6.1 Black-Scholes Comparison

Jump-Diffusion models inherently contain more volatility than the Black-Scholes model when the standard Brownian motion parameters are held constant. This is because Jump-Diffusion models combine the baseline volatility of Brownian motion with the additional volatility introduced by jumps. As a result, the Merton model produces higher option prices compared to the Black-Scholes model.

In the left plot of Figure 4, the Merton model uses the same diffusion parameter ($\sigma = 0.5$) as the Black-Scholes model. However, the additional jump-induced volatility inflates the Merton model option prices, making them consistently higher across all asset prices, S_t . Figure 5 plots the Δ values for the same models as Figure 4 across asset prices. The left plot helps emphasize that with constant diffusion parameters, the Merton model option prices begin increasing faster at lower asset prices, reflecting the aforementioned inflation. This highlights the impact of jump terms on overall pricing.

To quantify this difference, the t -period standard deviations of log-returns for the Black-Scholes and Merton models can be derived as [Mat04]:

$$SD_{BS}(S_t) = \sigma_{BS} \sqrt{t} \leq \sqrt{(\sigma_{Merton}^2 + \lambda\delta^2 + \lambda\mu^2)t} = SD_{Merton}(S_t)$$

Since larger volatility naturally translates to larger option prices in the Black-Scholes model, Merton model prices will always be greater than or equal to Black-Scholes prices when $\sigma_{BS} = \sigma_{Merton}$. Therefore, to bring the Merton model's pricing in line with the Black-Scholes framework, the diffusion parameter (σ_{Merton}) can be adjusted to account for the contribution of jump volatility.

$$\sigma_{Merton}^2 \leq \sigma_{BS}^2 - \lambda\delta^2 - \lambda\mu^2$$

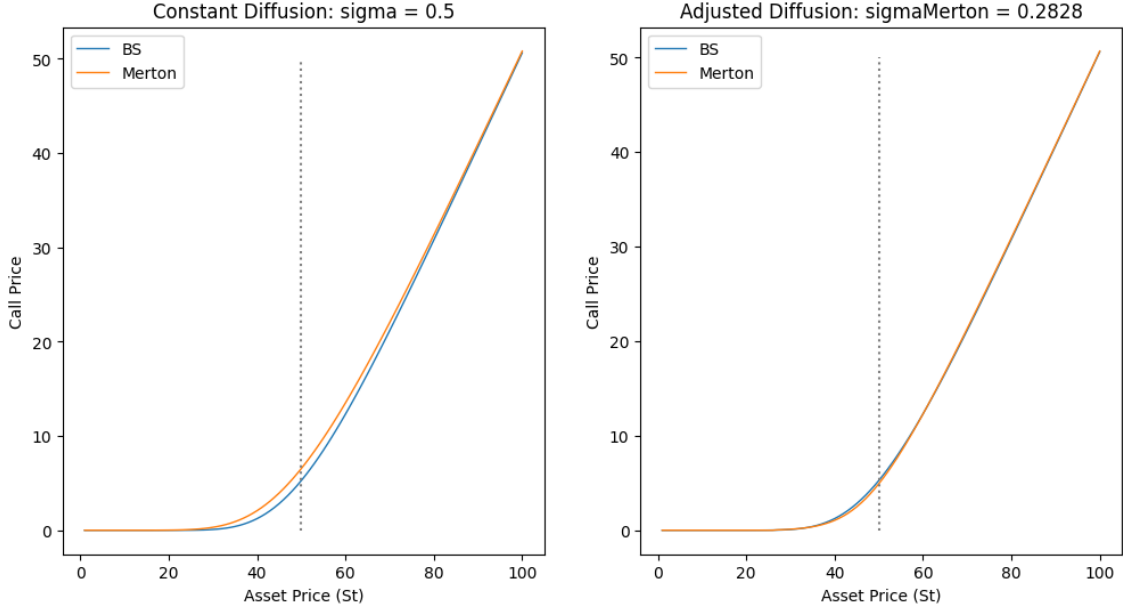


Figure 4: Black-Scholes and Merton model European call option pricing formulas. Plotted over asset prices (S_t) with constant parameters $K = 50, r = 0.05, T = 0.25, \lambda = 4, m = -0.05, \delta = 0.2, \sigma_{BS} = 0.5$ while varying $\sigma_{Merton} = 0.5$ on the left and $\sigma_{Merton} = 0.2828$ on the right.

The right plot of Figure 4 reflects this new pricing plot with an adjusted Merton model diffusion parameter of $\sigma_{Merton} = 0.2828$ when $\sigma_{BS} = 0.5$. Intuitively, if the true data-generating process (DGP) includes a jump term, the Black-Scholes model attempts to capture this volatility through its diffusion parameter (σ_{BS}). Consequently, when explicitly incorporating jumps in the Merton model, the diffusion parameter must be deflated by the amount of volatility attributed to the jumps. This adjustment ensures a balanced comparison between the two models and highlights the flexibility of Jump-Diffusion models in capturing the complexities of real-world market dynamics.

7 Conclusion

7.1 Report Summary

Throughout this report, the fundamental inspiration, methodology, and functionality of Jump-Diffusion models were explored. The challenges that inspired the development of Jump-Diffusion models were reviewed, illuminating some limitations of the Black-Scholes model. Compound Poisson Processes were defined and it was intuitively explained as to how they are used to model jumps, random changes in the underlying asset price. Jump-Diffusion models were then generally defined and two key models were simulated beside the Black-Scholes model to examine how they function in practice. Reading into the intuition behind the jumps as a representation of investors' sentiment inspired further study. A brief look at option pricing exposed the similarities of the derivations to those of their predecessor, the Black-Scholes model. Finally, plotting Merton model option pricing next to the Black-Scholes formula provided an again enhanced understanding of the mechanisms of the jumps as a tool for modelling in financial markets.

7.2 Questions of Interest

What are Jump-Diffusion models and how do they work?

In summary, they comprise an attempt to address shortcomings of the Black-Scholes model via modelling investor sentiment, or external market information, as jumps. Specifically, they are additive extensions

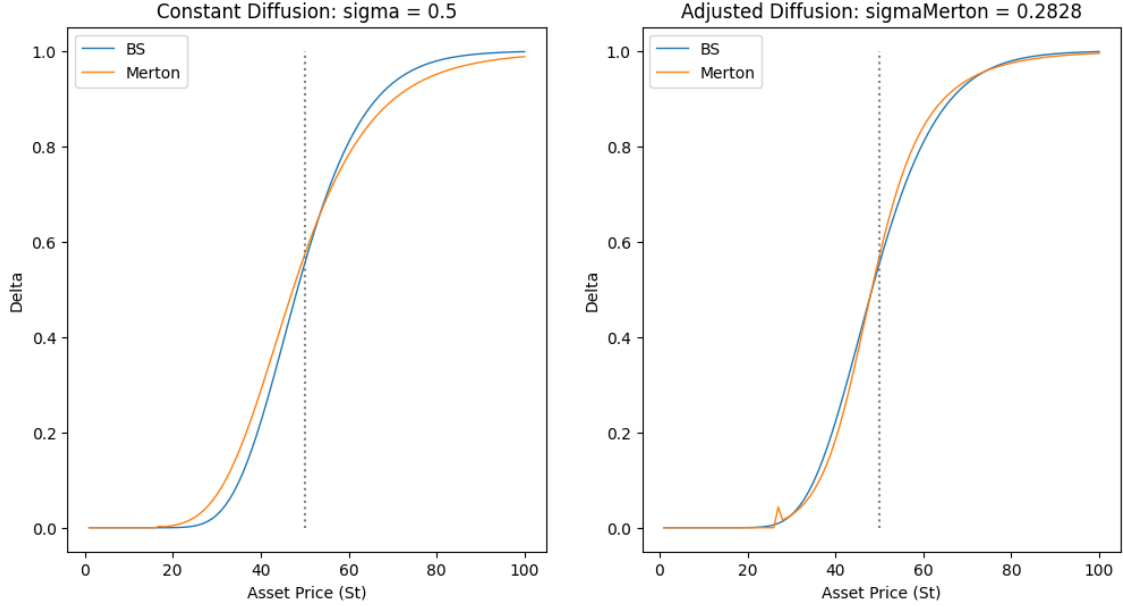


Figure 5: Black-Scholes and Merton model European call option pricing $\Delta = \frac{\partial f}{\partial S}$. Plotted over asset prices (S_t) with constant parameters $K = 50, r = 0.05, T = 0.25, \lambda = 4, m = -0.05, \delta = 0.2, \sigma_{BS} = 0.5$ while varying $\sigma_{Merton} = 0.5$ on the left and $\sigma_{Merton} = 0.2828$ on the right.

of the Black-Scholes model using compound Poisson processes to induce jumps with variable quantities, arrival times, and sizes. The introduction of jumps results in financial asset returns that follow a leptokurtic distribution, in turn reflecting a volatility smile. Due to the similarity to the Black-Scholes model, many of the same derivation techniques can be applied, becoming more complicated depending on the choice of jump size distribution, e.g. Merton vs. Kou models.

How do they compare to the Black-Scholes model?

The Black-Scholes model is rightfully appreciated for its simplicity and versatility, making it easy to understand and less prone to significant misrepresentations of market conditions. However, it leaves many financial asset price features unaddressed. On the other hand, Jump-Diffusion models address two of these features, the Leptokurtic distribution of asset returns and the additional pricing capability of the volatility smile. Further, they remain fairly simple in nature while attempting to model external market information. Of course, these advantages come with drawbacks. Adding any additional terms introduces at least some degree of complexity, this becomes especially evident when pricing options. In practice, the additional parameters induced by the jump component also require tuning. Finally, despite addressing two more features of financial asset prices, Jump-Diffusion models still leave many unaddressed, e.g. Volatility clustering.

7.3 Future Investigations

Today, there is a large and growing body of research into Jump-Diffusion models, their extensions, and applications. Understandably, the models tend to perform best in markets and other settings that are naturally observed to contain sporadic jumps. Some examples include credit risk modelling where credit events can lead to drastic changes, insurance claims and losses, and more. Like the original intentions behind Jump-Diffusion models, extensions, combinations, and other variations of jump process models continue to be explored to better reflect other features unaddressed by the Black-Scholes model. One such example would be the combination of Jump-Diffusion models with stochastic volatility models to tackle volatility clustering on top of the leptokurtic distribution and volatility smile.

Following this report, a natural option for future investigation would be the Kou model's European option

pricing formula, followed by the path-dependent option pricing formulas that inspired its development. Further, the derivations of the Merton model option pricing formula were summarized. The details behind Dr. Merton's choice of risk-neutral measure and the adjustments taken by Dr. Kou are worth investigating to develop a strong understanding of Jump-Diffusion model mechanics.

References

- [CT04] Rama Cont and Peter Tankov. *Financial modelling with jump processes*. Financial Mathematic Series. London, UK: Chapman Hall/CRC, 2004.
- [Kou07] S.G. Kou. *Chapter 2 Jump-Diffusion Models for Asset Pricing in Financial Engineering*. Vol. 15. Handbooks in Operations Research and Management Science. Elsevier, 2007, pp. 73–116.
- [Mat04] K. Matsuda. “Introduction to Merton Jump Diffusion Model”. In: (2004).